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An integrated paleomagnetic study of Rio Grande de Santiago volcanic succession (trans-Mexican volcanic belt): revisited

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Abstract

We carried out an integrated paleomagnetic, rock-magnetic and paleointensity study of Miocene volcanic succession from the trans-Mexican volcanic belt (TMVB) north of Guadalajara. A total of 37 consecutive basaltic lava flows (326 oriented standard paleomagnetic cores) were collected at Lazo locality. Continuous susceptibility measurements with temperature and hysteresis experiments yield in most cases reasonably reversible curves with Curie points close to that of pseudo-single-domain magnetite. Two geomagnetic reversals were observed in the 300 m thick composite section. Paleosecular variation was lower than the one observed in general during Miocene. It appears that the volcanic units have been emplaced during a relatively short time span of about 1 Ma. The mean paleomagnetic directions obtained from this study do not differ significantly from that expected for the middle Miocene. The mean paleomagnetic direction calculated from all data is $I = 31.1^\circ$, $D = 354.6^\circ$, $k = 124$ and $\alpha_{95} = 2.1^\circ$, $N = 37$. Seventy-two samples with apparently preserved primary magnetic mineralogy and without secondary magnetization, mostly belonging to reverse polarity chron were pre-selected for Thellier paleointensity determination. The flow-mean paleointensity values are ranging from 22.4 ± 3.4 to $53.8 \pm 6.0\,\mu$T and the corresponding virtual dipole moments (VDMs) are ranging from (5.4 ± 0.8) to $(12.0 \pm 1.4) \times 10^{22}$ A m^2. This corresponds to mean value of $(7.7 \pm 2.2) \times 10^{22}$ A m^2, which is close to present day geomagnetic field strength. Altogether, our data suggest the existence of relatively high geomagnetic field strength undergoing low fluctuations. © 2002 Elsevier Science B.V. All rights reserved.

Keywords: Paleointensity study; Paleomagnetic study; Rock-magnetic study

1. Introduction

The trans-Mexican volcanic belt (TMVB) is one of the largest continental volcanic arcs of the North American plate. It spans about 1000 km from the Pacific to the Gulf of Mexico (Fig. 1). Despite the abundance of thick lava sequences with quite high extrusion rates, the TMVB have been relatively little studied from a paleomagnetic point of view. Previous studies (e.g. Urrutia-Fucugauchi and Böhnel, 1988; Urrutia-Fucugauchi and Rosas-Elguera, 1994; Alva-Valdivia et al., 2000) were aimed for tectonic evolution of central Mexico rather than documenting fluctuations of Earth's magnetic field in terms of both directions and intensity. Older studies,

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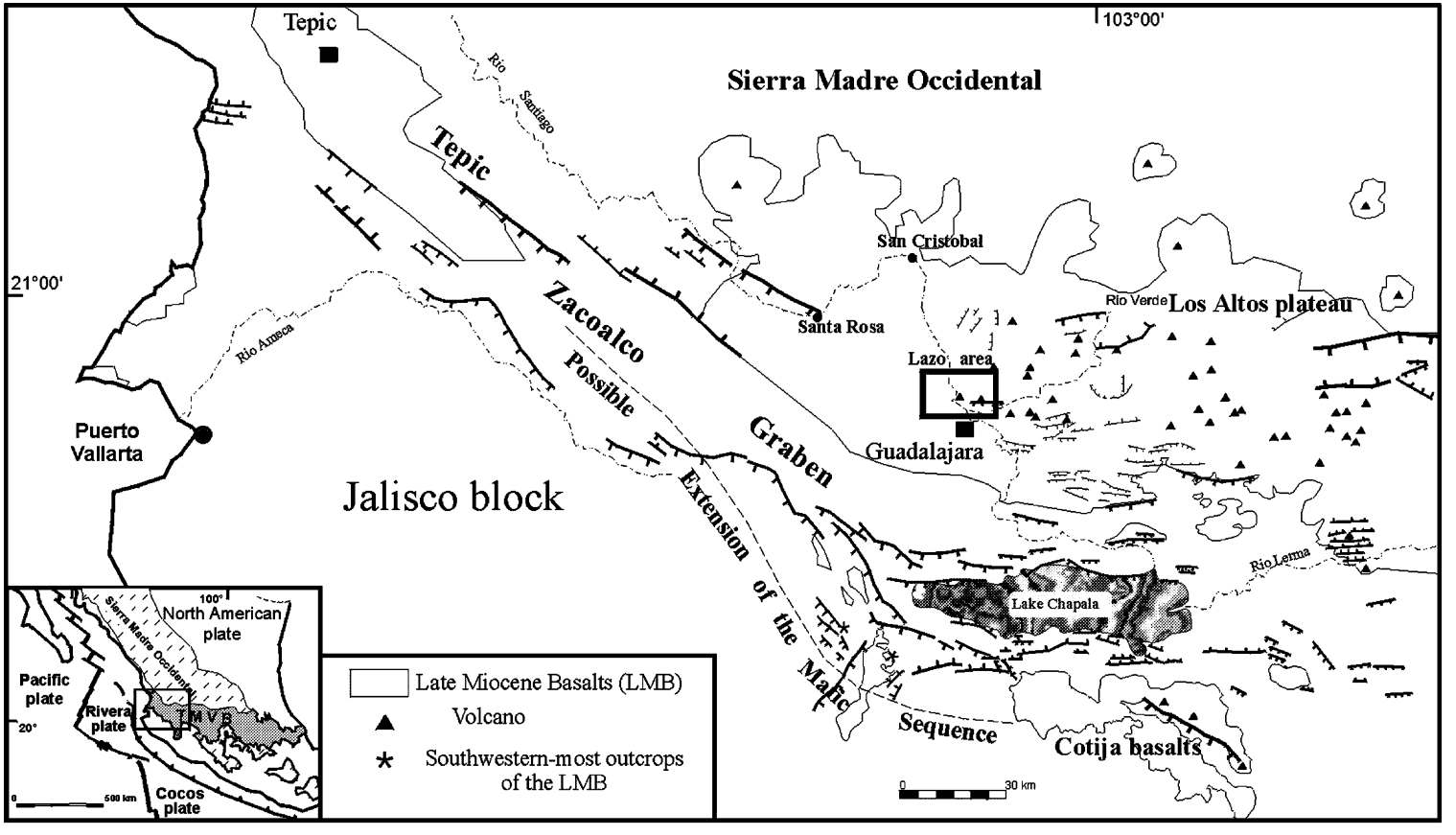


Fig. 1. Simplified geologic map of Late Miocene mafic lavas (in grey) and main fault systems of western trans-Mexican volcanic belt.

obtained with methods that have been since superseded, may not be entirely reliable and most of them do not satisfy basic selection criteria to be included in currently available paleomagnetic databases (Johnson and Constable, 1996; McEllhinny and McFadden, 1997; see also IAGA paleomagnetic database at http://www.ngu.no/dragon/Palmag/paleomag.htm). For this reason, IAGA News (1997) resolution recommends that attention should be given to repeating early studies on important sections using modern field and laboratory techniques.

Watkins et al. (1971) conducted one of the early studies in the TMVB, and reported paleomagnetic results from Rio Grande de Santiago lava flows. They studied four middle Miocene volcanic units and both normal and reverse magnetic polarities were observed. Let us note that no detailed magnetic cleanings were performed on these samples. The small number of studied sites may be mainly explained by lack of accessibility to the outcrops. In early 1990s, Mexican electricity agency Comision Federal de Electricidad (CFE) built the high quality road from north of Guadalajara to Rio Grande Santiago level at village Lazo. This allowed to access to huge volcanic sequences constituted by 37 independent consecutive lava flows, also named as San Cristobal basalts (Watkins et al., 1971). In present work, we have studied in details this single sequence characterized by unambiguous stratigraphy, so that sampling artifacts due to imperfect correlation and possible overlaps are excluded. One of the aims of present study was to supercede the data of Watkins et al. (1971) to be not robust and of low technical quality. In this study a considerable effort was spent on detailed rock-magnetic investigations to identify magnetic carriers, responsible for remanent magnetization and their paleomagnetic stability. All 326 samples were stepwise demagnetized by both alternating field (AF) and thermal procedures, which allowed precise determination of flow-mean paleodirections for all sites. Results are used to better constraint the tectonics of study area and to document the paleosecular variation of geomagnetic field during the middle Miocene. Moreover, 72 samples were pre-selected for Thellier paleointensity experiments. Altogether 41 samples coming from 11 lava flows yielded reliable absolute intensity estimates, which may contribute to our knowledge of geomagnetic field strength during Miocene and thus the transition mode between low intensity period during Mesozoic (so-called Mesozoic dipole low, Prévot et al., 1990) and the Neogene 'high' fields.

2. Geological setting and sampling details

The oldest volcanism related to the TMVB is a widespread basaltic succession, exposed from the Tepic and Puerto Vallarta areas (Fig. 1) to the longitude of Mexico City (Ferrari et al., 2000; Alva-Valdivia et al., 2000). Around Guadalajara region this unit is found to the north and south of Lake Chapala, forming Los Altos plateau (Ferrari et al., 1994) and the Cotija basalts (Rosas-Elguera, unpublished data). To the north of Guadalajara, at Lazo area (Fig. 2), this volcanism is characterized by plateau-like structures. An estimated volume of 3000 km^3 gives an average eruption rate of 0.8 km^3/ky (Ferrari et al., 1994). Stratigraphic and radiometric studies have shown that these basaltic lavas were emplaced in a period between about 11 and 8.5 Ma along the Rio Santiago canyon (Nieto Obregon et al., 1981; Damon et al., 1979; Moore et al., 1994), but older K–Ar dates (12–13.5 Ma) have been reported for the Los Altos plateau (Verma et al., 1985; Castillo-Hernandez and Romero-Rios, 1991). To the south, at Lake Chapala and Cotija areas the lavas were erupted between 7.6 and 8.7 Ma (Rosas-Elguera, unpublished data).

The best exposures of the Miocene basalts can be seen along the Rio Santiago canyon. Between Santa Rosa and San Cristobal more than 600 m of basalts are exposed (Moore et al., 1994) and at the Lazo section, 37 consecutive basaltic lava flows give only 300 m (Figs. 2 and 3). The tectonic framework around Lazo area has been a subject of discussion. West of Lazo section, Nieto Obregon et al. (1981) proposed an active right strike-slip faulting, which was later supported by Allan et al. (1991), Garduño et al. (1993), and Moore et al. (1994). Detailed stratigraphic and structural studies however evidenced that strike-slip faulting only cut older rocks rather than the Late Miocene basalts and an extensional tectonics has dominated since Late Miocene to quaternary (Michaud et al., 1991; Quintero et al., 1992; Ferrari et al., 1994). To the east of Lazo area, a left-lateral transtension has been proposed for Late Miocene times (Urrutia-Fucugauchi and Rosas-Elguera, 1994;

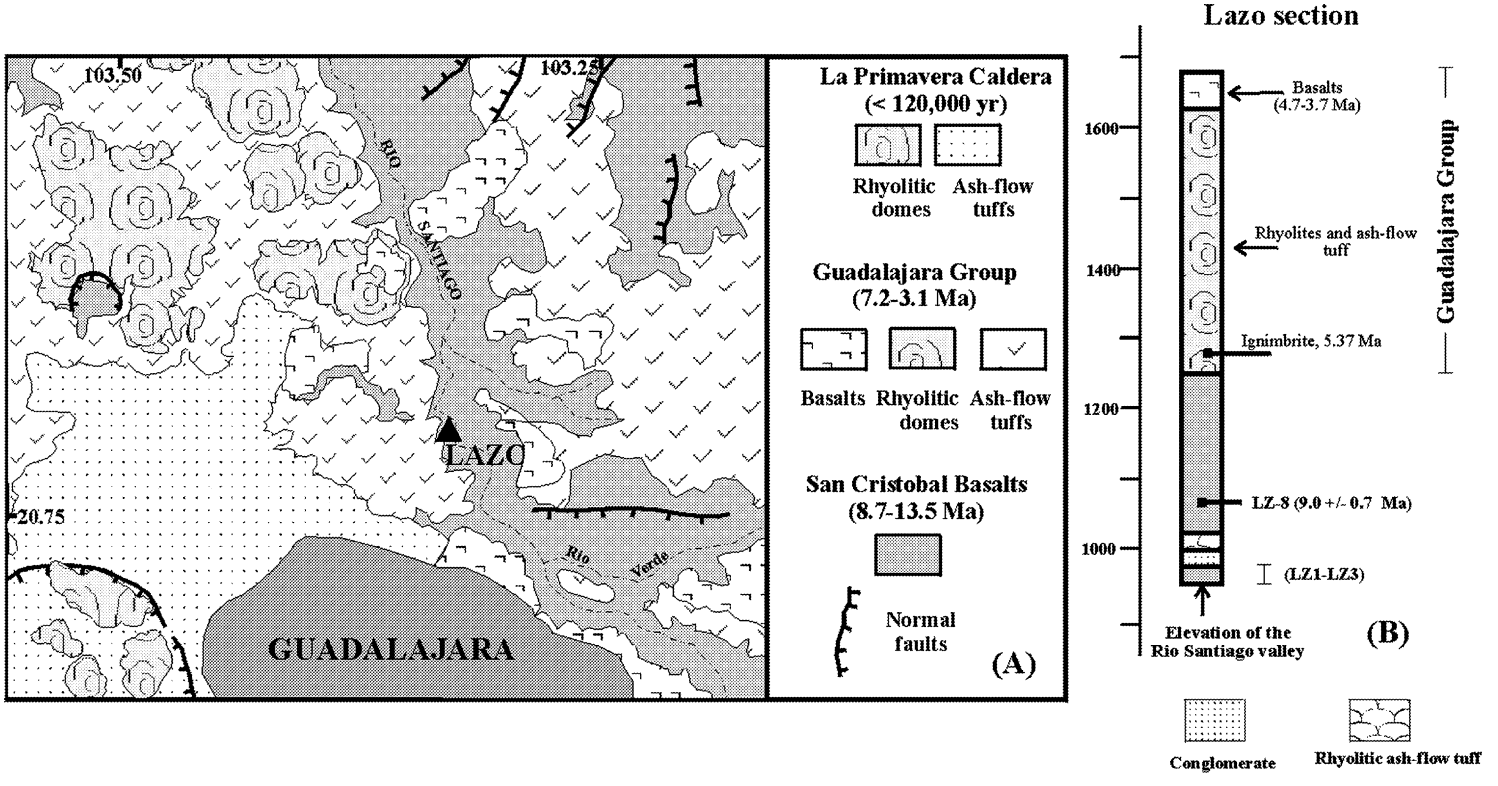


Fig. 2. Location (A) and relative stratigraphy (B) of Lazo section (see also text).

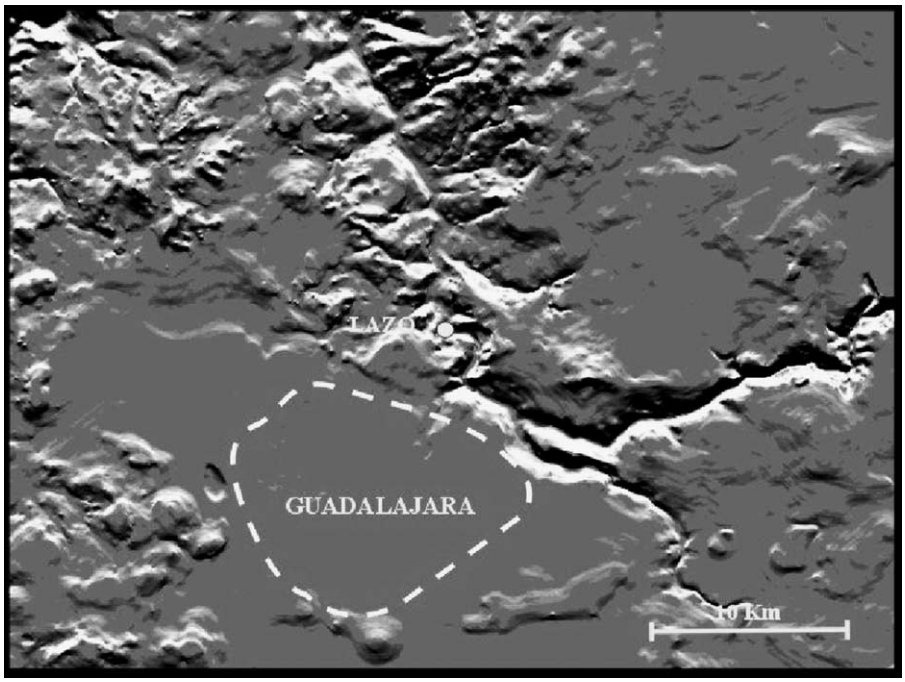


Fig. 3. A satellite image of surroundings of Guadalajara with indication of Lazo section.

Ferrari et al., 2000; Alva-Valdivia et al., 2000). Thus the Lazo area is a critical site to document the eruption rate of the Miocene basaltic succession and whether this succession has been affected by tectonic rotation around a vertical axis.

We sampled the Lazo section along the road from the CFE station to the village of Lazo. The geologic/geographic setting and relative stratigraphy of Lazo sub-horizontal stratified section is shown in Figs. 2 and 3. The site is located at 20°47′N latitude and 103°19′W longitude. The first three sites (LZ1, LZ2 and LZ3, Fig. 2) were sampled just below of conglomerate and ash-flow tuff level. Remaining 34 lava flows are all stratigraphically above of this level. Based on three concordant K–Ar ages, Watkins et al. (1971) proposed an age interval from 9.27 to 8.75 Ma, as the best estimate of the time of emplacement of the sequence. Additional K–Ar analyses we carried out yield basically the same results. Radiometric analyses on whole-rock samples belonging to site LZ8 (Fig. 2) were done at the Département de Minéralogie, Université de Genève (Switzerland). Age was calculated using the constants recommended by Steiger and Jäger (1977). Samples were obtained by trituration with steel-jaws, sieving (250, 350 and 500 μm), selection of the best fraction (usually the 350–500 mm), washing and separation of a whole-rock without phenocrysts by magnetic methods (Isodynamic Frantz separator). The K content of each sample was measured by XRF on 50 mg aliquots using a specific regression for measuring K in K–Ar samples. Analytical precision was better than 2%. About 150 mg whole-rock was degassed under vacuum at, approximately 100 °C for 10–12 h before analysis in order to reduce atmospheric contamination. Argon was extracted by complete sample fusion in a double vacuum resistance furnace with a Ta crucible and a Mo liner. The obtained gasses were mixed with a known amount of ^{38}Ar spike and purified with liquid nitrogen and a SAES getter in a stainless steel extraction line. Measurements were done in static mode with an AEI MS-10 mass-spectrometer with a permanent magnet of 4.1 kG and connected to a computer for processing data. Analytical precision on ^{40}Ar and ^{38}Ar peak heights was more than 0.5%, and 1% on ^{36}Ar. The data was calibrated with an internal standard and

the international standards LP-6 and HD-B1 biotites. This experiment yielded an age of 9.0 ± 0.7, which is compatible with the dates reported by Watkins et al. (1971).

In total, 326 oriented samples belonging to 37 consecutive lava flows were collected. Commonly, the outcrops extend laterally over a few tens of meters and in these cases we drilled typically 8–11 standard paleomagnetic cores per flow. The samples were distributed throughout each flow both horizontally and vertically in order to minimize effects of block tilting and lightning. In general, samples were obtained at the very bottom of flows with the hope of collecting samples with the finest grained material. At a few, small outcrops, only seven samples were collected. Cores were obtained with a gasoline-powered portable drill, and then oriented with both magnetic and sun compasses.

3. Rock-magnetism

3.1. Viscosity index

To obtain reliable and precise measurements of the remanence during progressive thermal demagnetization it is recommended (Prévot et al., 1983) that magnetic viscosity is low. Determination of the viscosity index (Thellier and Thellier, 1944; Prévot et al., 1983) allows to estimate the capacity of a sample to acquire a viscous remanent magnetization, and is therefore useful to obtain information about their paleomagnetic stability. For this purpose, we placed the samples during 20 days with one of their axes aligned with Earth's magnetic field. After measuring their magnetization (M_d), they were placed for another 20 days in a field-free space, and the magnetization (M_0) was measured again. This allows to calculate the viscosity index $V = ((Z_d - Z_0):M_{NRM}) \times 100$, where Z_d and Z_0 are, respectively the magnetization components of M_d and M_0 parallel to the magnetizing field, M_{NRM} the intensity of natural remanent magnetization. All samples were subjected to these experiments and although viscosity indexes varied between 0 and 22.8, most values were lower than 5%. The higher values belong to sites LZ28, LZ29 and LZ30. Generally speaking, the studied samples show a relatively low capacity of acquiring viscous remanent magnetization.

3.2. Curie points and thermal stability of samples

Low-field susceptibility measurements (k–T curves) under air were carried out using a highmoore susceptibility bridge equipped with furnace in Mexico City laboratory. One sample from each site were heated up to about 600 °C (sometimes to 650 °C) at a heating rate of 10 °C/min and then cooled at the same rate. Curie temperature was determined by the Prévot et al. (1983) method.

Two different types of behavior were detected: 22 samples show the presence of a single magnetic/ferrimagnetic phase with Curie point compatible with relatively low-Ti titanomagnetite (Fig. 4, samples LZ7-035C, LZ6-031D and LZ1-003). However, in some cases, the cooling and heating curves are not perfectly reversible (sample LZ1-003). Microscopic observations on polished sections also show that the main magnetic mineral is low-Ti titanomagnetite associated with exsolved ilmenite, probably formed as a result of oxidation of titanomagnetite during initial cooling. Remained samples yield two different thermomagnetic phases during heating (Fig. 4; samples LZ14-072, LZ17-090 and LZ23-120). The lower 'Curie point' ranges between 300 and 420 °C, and the highest one varies between 550 and 580 °C. The cooling curve shows only a single phase, with a Curie temperature close to that of magnetite. Such irreversible k–T curves can be explained by titanomaghemite, which probably transformed into magnetite during heating (Özdemir, 1987).

3.3. Hysteresis and IRM measurements

Hysteresis measurements at room temperature were performed on all studied lava flows using the AGFM 'Micromag' apparatus of the paleomagnetic laboratory at Mexico City in fields up to 1.2 T. The saturation remanent magnetization (J_{rs}), the saturation magnetization (J_s) and coercitive force (H_c) were calculated after correction for the paramagnetic contribution. The coercivity of remanence (H_{cr}) was determined by applying progressively increasing backfield after saturation. Some typical hysteresis plots are reported in Fig. 5. The curves are symmetrical in almost all cases. Near the origin, no potbellied and wasp-waisted behaviors (Tauxe et al., 1996) were detected, which probably reflects very restricted ranges of the opaque mineral

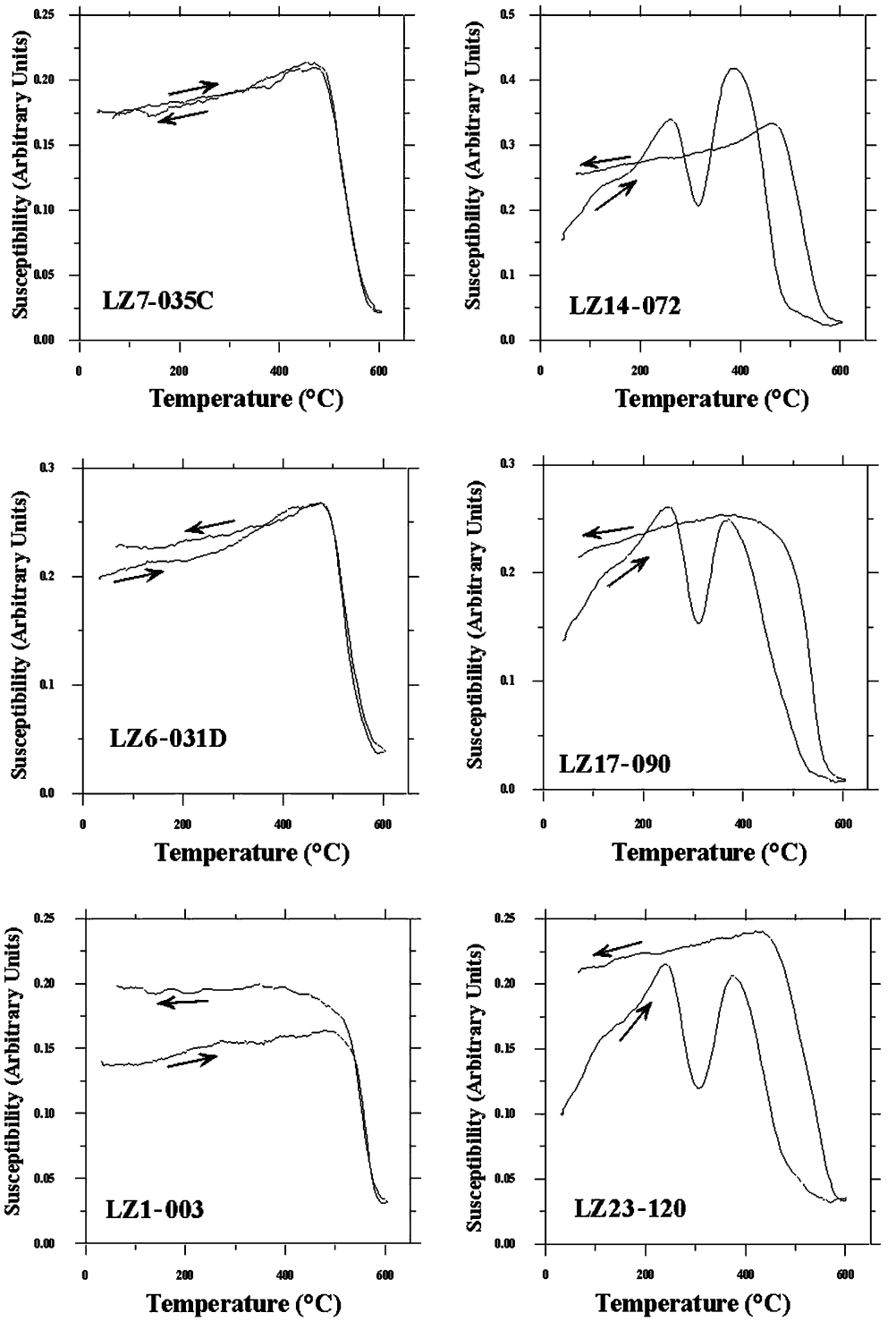


Fig. 4. Susceptibility vs. temperature (in air) curves of representative samples. The arrows indicate the heating and cooling curves.

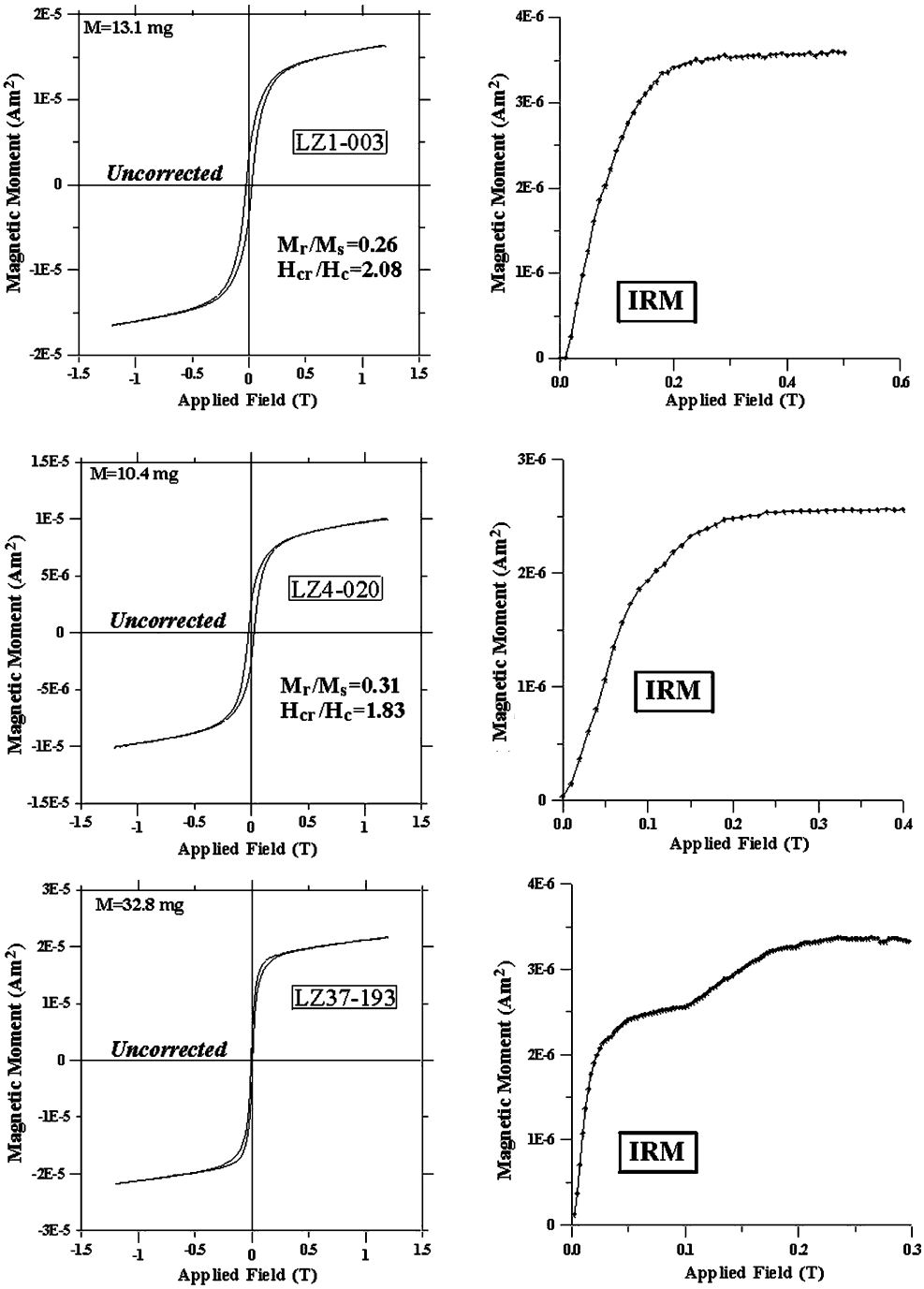


Fig. 5. Typical examples of hysteresis loops (uncorrected) and associated IRM acquisition curves of small chip samples from the studied volcanic units.

coercivities (samples LZ4-020 and LZ1-003). Only exception is site LZ37-193 (Fig. 5), which probably reflects the presence of ferrimagnetic phases with different coercivities. Most probably this behavior is due to the presence of dominantly single-domain and superparamagnetic grains (Goguitchaichvili and Prévot, 2000a). Except this single site, all samples seems to be contained 'small' pseudo-single-domain grains judging from hysteresis parameters values (Day et al., 1977). Let us note that if some superparamagnetic fraction exist also in these samples, the measured coercive force and saturation magnetization are somewhat lower or larger, respectively than those ferrimagnetic fractions alone. Corresponding isothermal remanence (IRM) acquisition curves (Fig. 5, right side) were found also very similar for all samples excepting site LZ37-193. Saturation is reached in moderate fields of the order of 100–200 mT, which points to some spinels as remanence carriers.

4. Remanence characteristics

4.1. Paleodirectional results

The remanent magnetization of 7–11 samples from each unit were measured with a JR-5A spinner magnetometer (sensitivity$\sim 10^{-9}$ A m^2). Measurements were recorded after stabilization of the remanence in this magnetometer. Both AF demagnetization using a laboratory made AF-demagnetizer and stepwise thermal demagnetization up to 570–610 °C using a MDT80 furnace (Magnetic Measurements Ltd.) were carried out. During thermal demagnetization, the low-field susceptibility at room temperature was measured after each step with a Bartington susceptibility meter.

In most of studied units a stable paleomagnetic component could be recognized (Fig. 6). Relatively small secondary components, probably of viscous origin, were easily removed applying 200 °C or 10 mT. Stronger secondary magnetization components were detected for seven sites (Fig. 6, samples LZ28-149B and LZ30-160A). The greater part of remanent magnetization, in most cases was removed at temperatures between 520 and 570 °C, which indicate, once again, to low-Ti titanomagnetites as responsible for magnetization. The median destructive fields (MDF) range mostly from 30 to 50 mT, suggesting pseudo-single-domain grains as remanent magnetization carriers (Dunlop and Özdemir, 1997).

A characteristic magnetization direction was determined by the least squares method (Kirschvink, 1980), five to nine points being taken in the principal component analysis for this determination. The obtained directions were averaged by unit and the statistical parameters calculated assuming a Fisherian distribution. The average unit directions are very precisely determined (Table 1). All α_{95} are less than 8.2°. This indicates the high quality paleomagnetic data.

4.2. Paleointensity determination

Following the paleodirectional and rock-magnetic results, altogether 72 samples belonging to 16 cooling units, yielding stable, one component magnetization with blocking temperatures compatible to Ti-poor titanomagnetite phase, low viscosity index (lower than 5%) and reasonably reversible *k–T* curves were selected for paleointensity experiments.

While procedures for determining the direction of the paleomagnetic field are now more or less standardized, there still exist significant inter-laboratory differences on how to best obtain reliable estimates of the paleomagnetic field intensity. The Thellier and Thellier technique and all derived approaches, among which Coe et al.'s (1978) modification received the largest audience in last years (Juàrez et al., 1998; Selkin and Tauxe, 2000; Goguitchaichvili et al., 2000b). This technique was used in present study to determine absolute geomagnetic paleointensity. The heatings and coolings were made in vacuum (better than 10^{-2} mbar) and the laboratory field set to 30 μT. Nine to ten temperature steps were distributed between room temperature and 560 °C. Temperature reproducibility between two heatings at the same step was in general better than 2°. The Thellier and Thellier method (Thellier and Thellier, 1959) allows one to check both stability of partial thermoremanent magnetization (pTRM) and natural remanent magnetization (NRM) left through control heatings. These TRM/NRM checks were performed after every second step throughout the whole experiment.

Paleointensity data are reported on the classical Arai–Nagata (Nagata et al., 1963) plot on Fig. 7 and results are given in Table 2. We accepted only

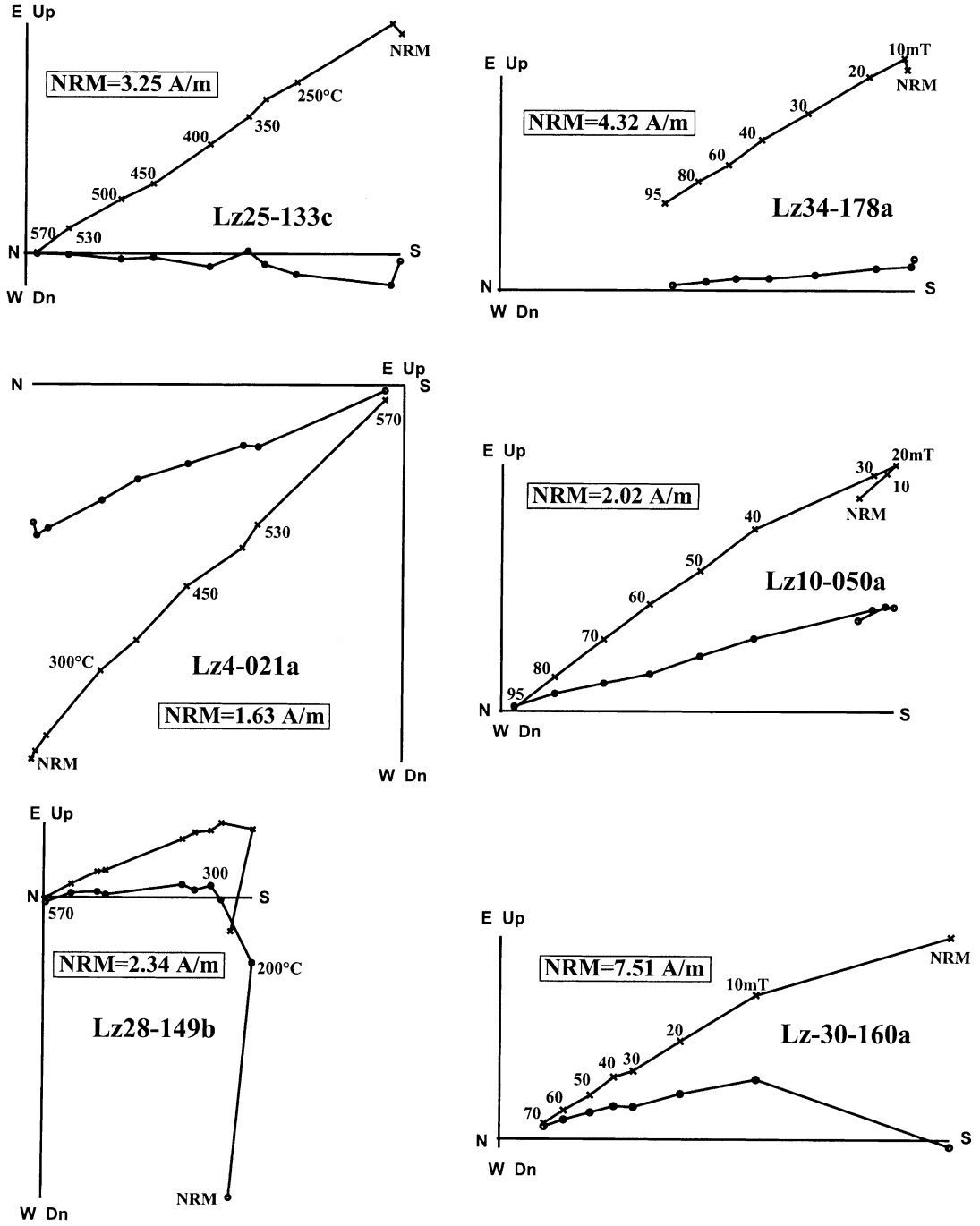


Fig. 6. Orthogonal vector plots of stepwise thermal or alternating field demagnetization of representative samples (stratigraphic coordinates). The numbers refer either to the temperatures (°C) or to the peak alternating fields (mT); (●) projections into the horizontal plane, (×) projections into the vertical plane.

Table 1
Flow-mean directions of cleaned remanence and corresponding VGP positions for Lazo section

Site	*n*/*N*	Inc	Dec	*k*	α_{95}	δ_D	P_{lat}	P_{long}	Pol
LZ37	8/11	−23.4	176.3	60.8	7.3	7.9	−81	280.5	R
LZ36	7/7	−21.9	171.5	123.5	5.5	9.6	−77.8	259.9	R
LZ35	7/8	−25	176.4	145.3	6.1	6.3	−81.9	252.4	R
LZ34	8/8	−27.3	181.1	257.3	3.5	6.8	−83.9	256.1	R
LZ33	8/8	−32.3	170.8	169.2	4.2	3.5	−80.9	329.6	R
LZ32	9/9	−27.2	165.3	143.1	4.3	9.2	−74.7	325.7	R
LZ31	8/8	−27.1	176.2	90.2	5.8	4.2	−82.9	257.9	R
LZ30	8/8	−28.6	169.4	60.7	7.2	5.2	−78.6	320	R
LZ29	8/8	−26.6	174.4	72.6	6.5	4.5	−81.6	297.4	R
LZ28	9/9	−30.3	176.8	112.8	4.9	2.1	−84.8	275.2	R
LZ27	8/8	−30.9	169.2	168	4.3	4.6	−79.1	328.1	R
LZ26	11/11	−26.6	172.5	135.4	3.9	4.9	−80.4	306	R
LZ25	8/8	−28.4	174.3	127.4	4.8	2.7	−82.4	302.9	R
LZ24	8/8	−28.2	177.9	173.3	4.2	4.1	−84.2	277.1	R
LZ23	9/9	−25.5	173.7	109.9	4.9	5.7	−80.7	298.2	R
LZ22	10/10	−25.2	170.5	179.8	3.6	6.8	−78.4	309.7	R
LZ21	9/9	−32.1	173.9	155.6	4.1	1.2	−83.5	319.7	R
LZ20	8/8	−28.4	187.7	132.5	4.8	12	−80.9	201.8	R
LZ19	8/8	−27.1	178.6	517.8	2.4	5.3	−83.7	269.3	R
LZ18	10/10	−32.3	174.5	412.5	2.5	1.2	−84	318.1	R
LZ17	9/9	−30.8	175	264.7	3.2	0.5	−83.9	308.2	R
LZ16	10/10	−32.6	168.2	152	3.9	5.7	−78.5	334.9	R
LZ15	7/7	−29.2	174.5	57.2	8.1	1.9	−82.9	304.6	R
LZ14	10/10	−28.6	178.6	99.2	4.9	4.3	−84.6	271.3	R
LZ13	10/10	−29	168.1	152.3	3.9	6	−77.6	324.9	R
LZ12	9/9	−35.8	184.6	76.3	5.9	9.6	−85.6	174.8	R
LZ11	9/9	−32.8	176.4	204.9	3.6	2.3	−85.7	309.6	R
LZ10	9/9	−31.5	177.2	155.4	4.1	2.3	−85.6	294.7	R
LZ9	8/9	−30.5	179.5	116.6	5.4	4.3	−85.9	265.5	R
LZ8	9/9	−25.2	183.2	271.4	3.1	9.5	−82.1	223.4	R
LZ7	7/7	42.8	353.6	56.2	8.2	11	82.7	204.3	N
LZ6	8/8	−46.4	178.5	678	2.1	16	−82.7	66.3	R
LZ5	11/11	41.1	357.3	111	4.3	10	86	218.1	N
LZ4	8/8	40.2	350.4	51.3	7.8	9.7	81.2	184.3	N
LZ3	10/10	39.4	349.6	120.2	4.4	9.3	80.1	179.4	N
LZ2	8/8	41.6	347.5	68.8	6.7	12	77.9	185.7	N
LZ1	8/8	35.7	342.8	67.1	6.8	11	73.8	167.2	N

N, the number of treated samples; *n*, number of specimens used for calculation; Inc, inclination; Dec, declination; *k* and α_{95}, precision parameter and radius of 95% confidence cone of Fisher statistics; δ_D, angular distance from the normal dipole direction; P_{lat} and P_{long}, latitude and longitude of VGP position; Pol, magnetic polarity.

determinations: (1) which were obtained from at least 5 NRM/TRM points corresponding to a NRM fraction larger than 1/3 (Table 2), (2) yielding quality factor (Coe et al., 1978) of about 5 or more, and (3) with positive 'pTRM' checks. In few cases, we accepted the individual determination with slightly lower quality factors (Table 2). However, paleointensity estimate from these samples are close to the site mean. In most cases the linearity was observed up to 540 °C (Fig. 7, sample LZ34-180C) and the control heatings were successful, i.e. the deviation of 'pTRM' checks were less than 15%. The direction of NRM left at each step, obtained from the paleointensity experiments are reasonably linear and point to the origin (Fig. 7, right side). No deviation of NRM left directions towards the direction of applied laboratory field was observed. A typical 'concave-up' behavior (Dunlop and Özdemir, 1997) was detected in some cases (Fig. 7,

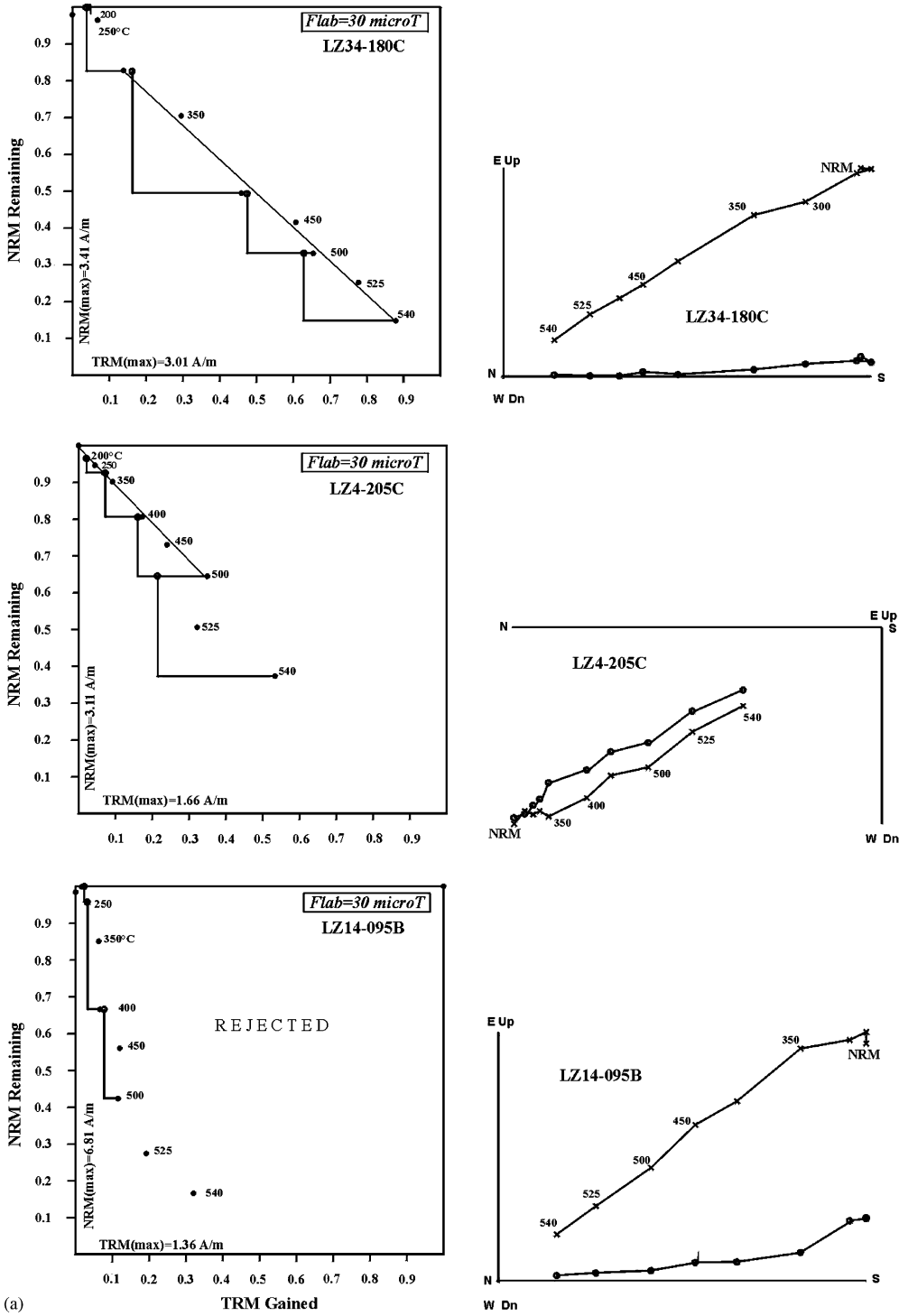


Fig. 7. The representative NRM/TRM plots and associated orthogonal diagrams from Rio Grande de Santiago samples. In the orthogonal diagrams we used same notations as in Fig. 6.

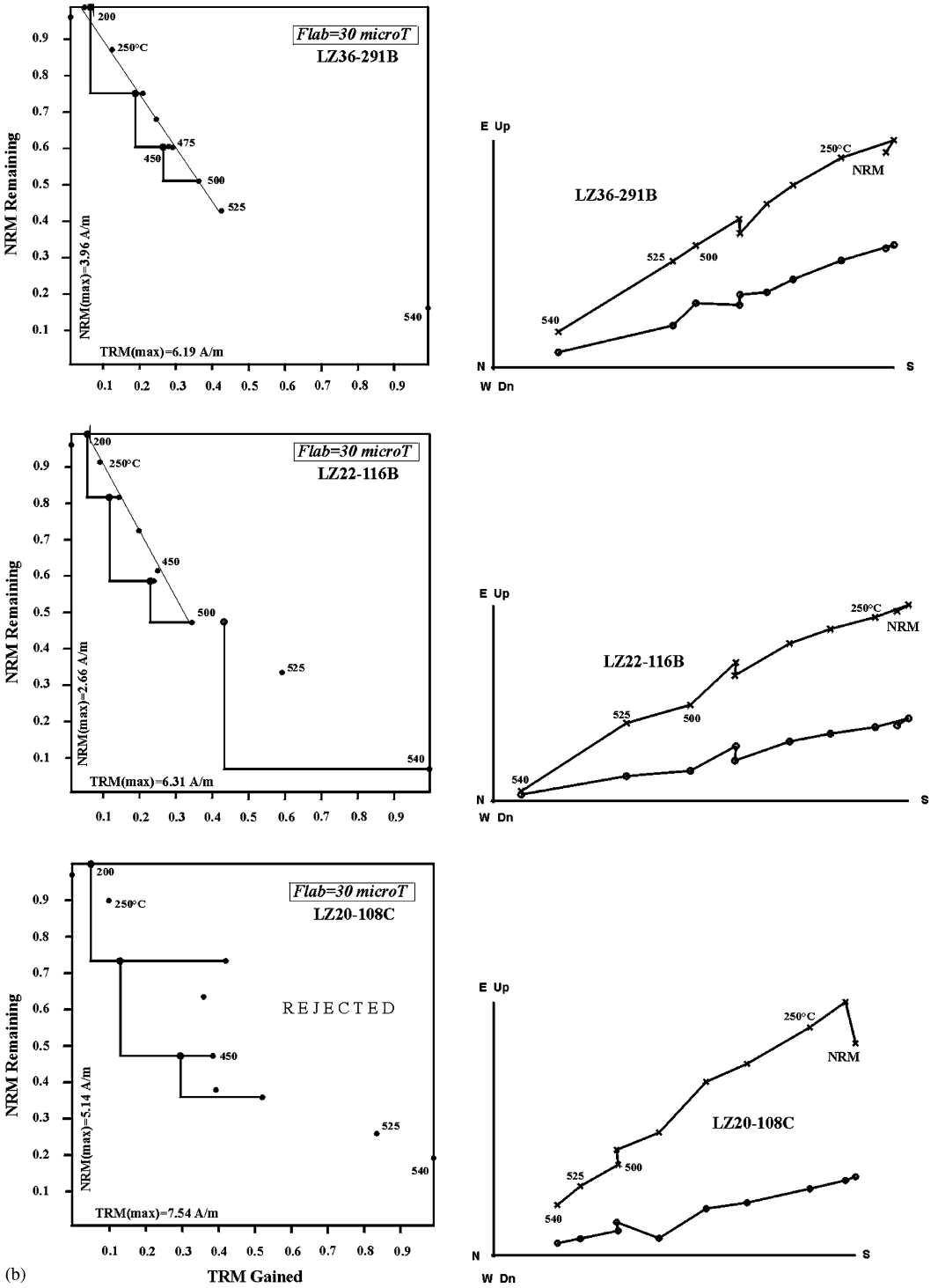


Fig. 7. (*Continued*).

Table 2
Paleointensity results from Rio Grande de Santiago volcanic lava flows

Site	Sample	n	T_{min}–T_{max}	f	g	q	$F_E \pm \sigma(F_E)$	VDM	$F_E \pm$ S.D.	VDMe
LZ1	99G001B	9	200–540	0.58	0.85	7.6	57.3 ± 3.7	12.8	53.8 ± 6.0	12.0 ± 1.4
	99G198C	8	200–520	0.49	0.82	5.9	57.2 ± 4.3	12.8		
	99G199C	7	250–520	0.57	0.76	4.9	46.8 ± 4.5	10.4		
LZ4	99G018B	8	250–540	0,57	0.74	11	35.0 ± 1.4	7.5	40.0 ± 6.1	8.6 ± 1.3
	99G206B	7	300–540	0.62	0.73	8.5	43.4 ± 2.3	9.3		
	99G207C	6	350–540	0.61	0.68	8.1	46.7 ± 2.2	10		
	99G205C	8	20–500	0.36	0.79	7.3	34.7 ± 1.2	7.4		
LZ5	99G022B	9	200–540	0.68	0.74	9.6	48.9 ± 2.5	10.4	43.0 ± 5.4	9.2 ± 1.2
	99G023B	8	200–520	0.41	0.83	5.8	47.1 ± 2.6	10		
	99G027B	8	250–540	0.58	0.78	15	41.3 ± 1.2	8.8		
	99G208C	7	300–540	0.83	0.72	10	35.1 ± 1.7	7.5		
	99G210B	8	200–520	0.44	0.83	4.4	42.8 ± 3.6	9.1		
LZ6	99G028A	7	250–520	0.36	0.83	6.1	31.2 ± 1.6	6.3	26.2 ± 5.8	5.3 ± 1.2
	99G029B	7	250–520	0.4	0.76	7.5	33.7 ± 1.2	6.8		
	99G030B	7	250–540	0.41	0.68	6.3	23.7 ± 0.8	4.8		
	99G033B	7	250–520	0.51	0.64	6.4	21.3 ± 1.1	4.3		
	99G212C	8	250–540	0.57	0.72	5.5	21.2 ± 3.5	4.3		
LZ7	99G034A	6	350–540	0.51	0.73	5.4	34.2 ± 2.4	7.2	33.6 ± 6.4	7.0 ± 1.3
	99G035A	6	300–520	0.53	0.79	4.8	39.7 ± 4.3	8.3		
	99G036A	8	250–540	0.57	0.72	7.6	26.9 ± 2.1	5.6		
LZ9	99G043B	8	250–540	0.47	0.84	6.4	31.6 ± 2.4	7.3	26.2 ± 3.8	6.1 ± 0.8
	99G044B	8	250–540	0.57	0.88	4.7	27.9 ± 3.1	6.5		
	99G045A	7	300–540	0.58	0.75	6.8	25.6 ± 1.6	5.9		
	99G046B	6	350–540	0.72	0.7	7.4	21.5 ± 1.1	4.9		
	99G220C	6	300–520	0.38	0.73	4.4	24.5 ± 2.2	5.7		
LZ21	99G109A	6	300–520	0.47	0.74	9.2	44.6 ± 0.8	10.2	48.5 ± 5.5	11.1 ± 1.3
	99G251B	6	250–500	0.44	0.8	6.9	52.4 ± 2.7	11.9		
LZ22	99G113B	6	250–500	0.49	0.76	14	36.6 ± 0.5	8.8	29.9 ± 4.9	7.2 ± 1.2
	99G114B	6	250–500	0.51	0.79	5.5	27.5 ± 2.3	6.6		
	99G116B	7	200–500	0.5	0.79	5.7	25.3 ± 1.5	6.1		
	99G254C	5	300–500	0.55	0.75	4.6	30.6 ± 2.9	7.4		
LZ34	99G176B	8	300–560	0.78	0.83	12	43.8 ± 2.3	10.4	37.8 ± 8.8	9.0 ± 2.1
	99G177A	7	300–540	0.65	0.77	7.6	43.5 ± 4.3	10.3		
	99G178B	8	300–560	0.65	0.78	8.1	45.2 ± 2.9	10.7		
	99G180C	7	300–540	0.64	0.79	14.1	27.5 ± 2.8	6.5		
	99G287C	7	300–540	0.53	0.79	6.2	28.9 ± 2.1	6.9		
LZ36	99G291B	8	200–520	0.47	0.79	5.7	30.2 ± 2.3	7.4	28.4 ± 2.6	6.9 ± 0.7
	99G292B	6	300–520	0.46	0.78	4.8	26.5 ± 1.9	6.4		
LZ37	99G295A	6	350–560	0.78	0.74	15	22.1 ± 0.8	5.4	22.4 ± 3.4	5.4 ± 0.8
	99G296A	7	300–540	0.68	0.69	6.9	25.9 ± 1.5	6.3		
	99G294B	7	300–540	0.62	0.67	6.7	19.2 ± 1.2	4.7		

n, the number of NRM/TRM points used for palaeointensity determination; T_{min}–T_{max}, the temperature interval used; f, g and q, the fraction of extrapolated NRM used, the gap factor and quality factor (Coe et al., 1978), respectively; F_E, paleointensity estimate for individual specimen; $\sigma(F_E)$ standard error of F_E; F_E, the average paleointensity of individual lava flow, the plus and minus sign corresponding to standard deviation; VDM and VDMe, individual and average virtual dipole moments.

sample LZ14-095B). An important loss of NRM without any noticeable TRM acquisition but with positive pTRM checks is observed. This phenomenon can be due to irreversible variations of coercive force (Kosterov and Prévot, 1998) at low temperature and can be interpreted as transformation from a single-domain 'metastable' state to polydomain state which results in a large NRM lost without any correlated TRM acquisition during the subsequent cooling.

Finally, only 41 samples, coming from 11 individual basaltic lava flows, yielded acceptable paleointensity estimates (Table 2). For these samples the NRM fraction *f* used for determination ranges between 0.36 and 0.78 and the quality factor *q* from 4.4 to 15.3 being generally greater than 5.

5. Discussion and main results

5.1. Possible implications for tectonics and volcanic stratigraphy

We consider the characteristic paleomagnetic directions determined in this study to be of primary origin. This is confirmed by the fact that normal and reversed polarities were determined. In addition, thermomagnetic curves show that the remanence is carried in most cases by Ti-poor titanomagnetite, resulting of oxy-exsolution of original titanomagnetite during the initial flow cooling, which most probably indicates thermoremanent origin of a primary magnetization. Some samples seem to contain titanomaghemites as revealed by thermomagnetic analyses. It is possible that these samples carry chemical remanent magnetization. However, both experimental and theoretical studies (Heider and Dunlop, 1987; Özdemir and Dunlop, 1989; Nishitani and Kono, 1989) show that chemical remagnetization by magnetization have the same direction as the original TRM. Consequently, paleodirections were most probably unaffected by alteration and they can be used for tectonic and magnetostratigraphic application.

The mean paleodirection obtained in this study is $I = 31.1^\circ$, $D = 354.6^\circ$, $k = 124$, $\alpha_{95} = 2.1^\circ$, which corresponds to the mean paleomagnetic pole position $P_{\mathrm{lat}} = 84^\circ$, $P_{\mathrm{long}} = 129.8^\circ$, $k = 29$, $\alpha_{95} = 4.4^\circ$. These directions are in reasonably good agreement with the expected paleodirections for middle Miocene time (Fig. 8a) , as derived from reference poles given by Besse and Courtillot (1991) for North America (see also Alva-Valdivia et al., 2000). This probably suggests that no major tectonic deformation occurred in studied area since about 9 Ma. Our results are in disagreement with previous studies (Urrutia-Fucugauchi and Böhnel, 1988; Alva-Valdivia et al., 2000), which indicate some 15–20° anticlockwise tectonic rotations for the Rio Grande de Santiago canyon and surrounding areas, in accordance with the proposed left-lateral transtensional tectonic regime already proposed for this period. This controversy may be owned from the fact that previous studies are based on relatively limited number of sites.

With these results, when combined with single K–Ar datum, a direct correlation with the geomagnetic polarity time scale (GPTS) can be intended (Fig. 8b). Four magnetozones were identified at Lazo section. First five lava flows (LZ1–LZ5) are all normally magnetized and most probably were emplaced during chron C4Ar.2n. A quite well defined contact separates flows LZ5 and LZ6, although former yields reverse polarity magnetization which may tentatively assigned to chron C4Ar2r. Next flow (LZ7) is again normal and all remained lavas (in total 30 consecutive lava flows) are reverse, which may correspond to the magnetochron C4Ar1r at Cande and Kent (1995) reference GPTS. Based on our paleomagnetic and radiometric data, it may be speculated that the Rio Grande de Santiago volcanic units have been emplaced during a relatively short time span of 1 My or even less at Lazo locality.

5.2. Fluctuations of paleofield

A quite informative paleomagnetic record is obtained for the Rio Grande de Santiago Miocene volcanic succession (Fig. 9). Paleomagnetic data in volcanic units provide spot-readings of the evolution of the past geomagnetic field. Studies of the statistical dispersion of the paleomagnetic directions is the only way to describe long-term variations. Such studies of the paleosecular variations are numerous for the last 5 My. Very few data are however available for the Miocene period.

The classical formula $S_{\mathrm{F}}^2 = (S_{\mathrm{T}}^2 - S_{\mathrm{W}}^2)/n$ was used for estimating paleosecular variation in this study.

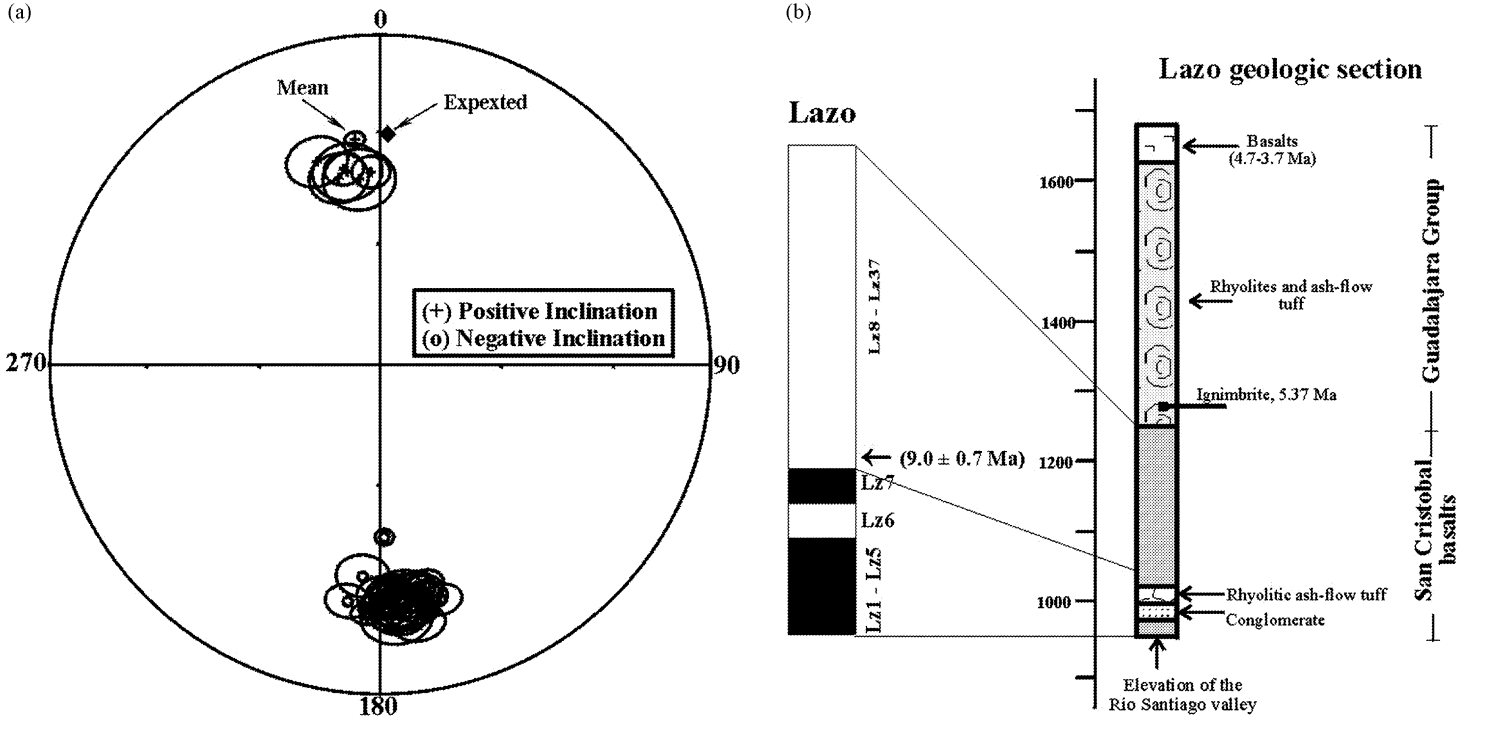


Fig. 8. (a) Equal area projections of the flow-mean characteristic paleodirections of Rio Grande de Santiago volcanics; (b) a relative magnetostratigraphy of Lazo section.

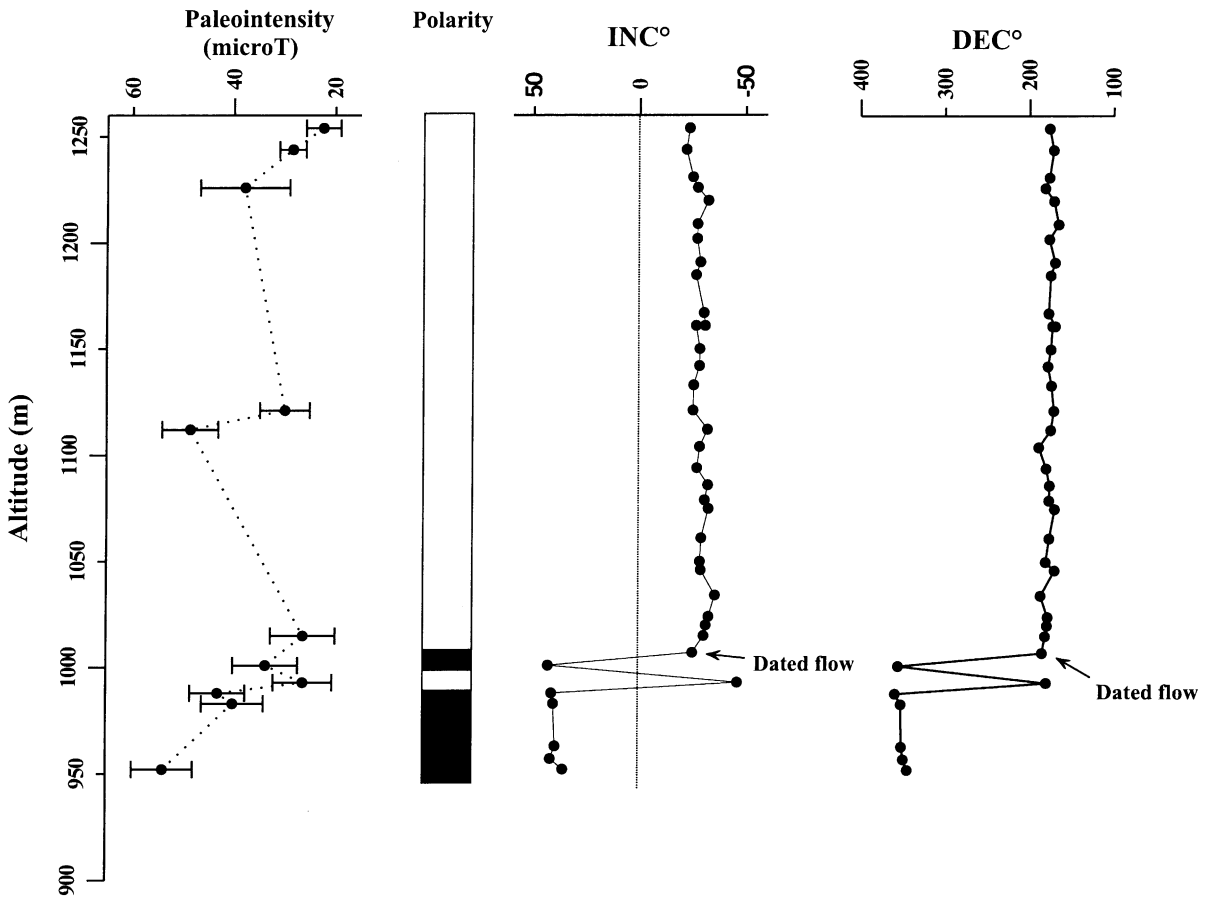


Fig. 9. Flow-mean paleoinetensity, inclination and declination for each flow of Lazo section against elevation.

Here, S_T is the total angular dispersion given by $S_T = \left[(1/N-1)\sum_{i=1}^{N}\delta i^2\right]^{1/2}$ (Cox, 1969), N the number of sites used in calculation, δi the angular distance of the ith virtual geomagnetic pole (VGP) from the axial dipole, S_W the within site dispersion, n the average number of samples per unit. McFadden et al. (1991) report a VGP scatter equal to 18.1 for the 20° latitude bands between 5 and 22.5 My. We obtained $S_F = 15.1$ (upper and lower limits: 17.9–13.1), which is lower (Fig. 10). We tried to compare VGP scatter obtained in this study to the dispersion obtained from the other Miocene volcanic sites, from America. There are only few studies directly available. In most cases, evidence for local tectonic rotations are reported and thus they cannot be correctly used for paleosecular variation study. The VGP changing from flow to flow found at Steens Mountain (Mankinen et al., 1985) volcanic succession (discarding intermediate polarity units) agree quite well with McFadden et al. (1991) model yielding VGP scatter equal to 18.9. The same calculation on Steens Mountain directional groups (Prévot et al., 1985) provide basically the same value. Up to now, there are very few studies reporting paleomagnetic results in Miocene volcanic rocks for the South American continent (Goguitchaichvili et al., 2000b). Most studies belong to the Andean Chain and it is thus difficult to separate local tectonic rotation from true paleosecular variation (Roperch et al., 1996).

Low paleosecular variation documented in this study may indicate to high eruption rate of volcanic

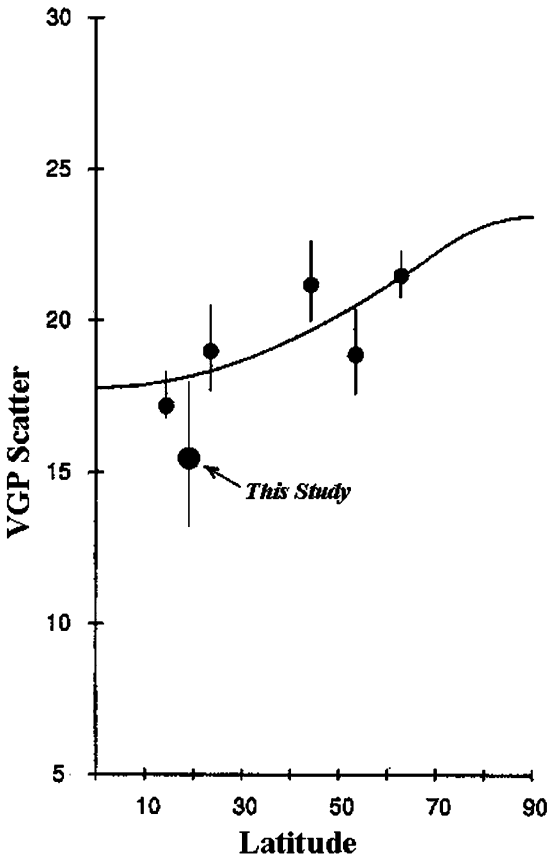


Fig. 10. Paleosecular variation of lavas (PSVL) for 5–22.5 Ma (retrieved from McFadden et al., 1991).

activity and thus relatively short emplacement time for the lava flows. Alternatively, these data support the hypothesis that the central Pacific region might have been a region of low non-dipole geomagnetic field in the Miocene as suggested by Doell and Cox (1971) for the Brunhes chron. Tanaka and Kono (1991) and Mankinen and Champion (1993) have also obtained data supporting the existence of a significant non-dipole field at central Pacific region. McEllhinny et al. (1996) have re-examined the old data from Hawaii and Tahiti and used a new statistical method to make proper account of those flows that have repeatedly sampled the same geomagnetic field vector. They concluded that the amplitude of the secular variation is consistent with values from other worldwide scattered sites, so that the hypothesis of the Pacific dipole window may be rejected.

5.3. Geomagnetic field strength during Miocene

Knowledge of the variation of geomagnetic paleointensity with geological time helps to understand how the geodynamo works. These data provide thus an important source of information on the physics of the Earth's deep interior. Despite recent growth of the quality and quantity of absolute geomagnetic intensity measurements reliable paleointensity data are still scarce and cannot be yet used to document the general characteristics of the Earth's magnetic field (Selkin and Tauxe, 2000). Because existing Miocene paleointensity data are scarce and of dissimilar quality (Riisager, 1999), new determinations are needed. These data may also help to know the transition mode between the Mesozoic low (Prévot et al., 1990) and the Neogene 'high' fields.

The difficulties related to absolute geomagnetic intensity determination on whole-rock extrusive volcanics are well illustrated in this study where only 41 samples out of 326 proved usable. The flow-mean paleointensity values obtained in this study range from 22.4 ± 3.4 to 53.8 ± 6.0 μT and the corresponding virtual dipole moments (VDMs) range from (5.4 ± 0.8) to $(12.0 \pm 1.4) \times 10^{22}$ A m^2. This corresponds to mean value of $(7.7 \pm 2.2) \times 10^{22}$ A m^2 (all data are used in calculation), which is close to present day geomagnetic field strength. Although our results are not numerous, some credit should be given because of good technical quality determination, attested by the reasonably high Coe et al.'s (1978) quality factors. However, there are not enough data to discuss VDM variation through time and it is necessary to combine our results with previously published estimates from the same period of time. In analyses of the Miocene paleointensity data, we considered only results obtained with the Thellier method for which positive pTRM checks attest the absence of alteration during heatings, at least three determinations per unit and an error of the mean paleointensity about 20% (Perrin and Shcherbakov, 1997). No transitional data is used in analyses. From IAGA paleointensity database (compiled by Mireille Perrin and Lisa Schnepp; Perrin et al., 1998), see also at ftp://dstu.univ-montp2.fr/pub/paleointb) only four studies (in total 19 determinations, i.e. obtained from 19 independent cooling units) fulfill these basic criteria (Table 3). Combination of Mexican paleoinetsnity data to comparable quality worldwide data yields a

Table 3
Selected average virtual dipole moments (10^{22} A m^2) for various regions during the Miocene (from Thellier palaeointensity data)

Region	Age	*N*	VDM	S.D.	Reference
Central France	8.6–9.7	3	3.9	2.7	Riisager, 1999
Oceanic basalts	6.4–10.3	2	4.8	4.2	Juàrez et al., 1998; Selkin and Tauxe, 2000
North America	~15.5	10	6.9	2.4	Prévot et al., 1985
Central Mexico	7.8–11	4	7.1	1.2	Goguitchaichvili et al., 2000c
TMVB	~9	9	7.7	2.2	This study

N refers to the number of cooling units used to calculate the average VDM (see also text).

mean VDM equal to $(6.1 \pm 1.4) \times 10^{22}$ A m^2, which is almost 80% of the present geomagnetic axial dipole (7.8×10^{22} A m^2 after Barton et al., 1996). Reliable paleointensity results for the Miocene are still scarce. Additional high quality absolute intensity determinations are needed to better constraint the geomagnetic field strength during the Miocene and the transition mode between Mesozoic low and the Neogene high field.

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